Retrieval that Cannot See the Future: Temporal Evaluation of Fact-Checked Claim Retrieval Project Proposal for NLP Course, Winter 2026/27 (revised after Phase 1 feedback; point-by-point answers in `rebuttal.pdf`)

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Abstract

Previously fact-checked claim retrieval (PFCR) ranks an archive of fact-checks for a social media post. Published benchmarks search the full archive, including fact-checks written after the post, while a fact-checker at post time can only search what already exists. We propose a *causal* evaluation protocol: for each post q with time t_q, the pool is $P(q) = \{f : t_f < t_q\}$, and we report Success@10 (fraction of queries with a gold fact-check in the top 10) and mean reciprocal rank (MRR) under random, size-matched random, and strict-causal pools on MultiClaim, with Polish as focus and English as control. Deliverables: the protocol and code that rebuilds the causal split from the restricted data (the authors asked us to keep id lists confidential); a results table of at least four systems under both protocols; and a gap decomposition with 50 hand-coded newly failing queries.

1 Scientific goal

Problem. PFCR matches a post q to earlier fact-checks f that already debunk the same claim (Shaar et al., 2020; Pikuliak et al., 2023). Standard evaluation lets q retrieve fact-checks published after t_q, which rewards matching a later article that quotes the post. We ask how much reported performance survives a time-respecting index, and why it drops.

Research questions. RQ1: How much Success@10 (S@10) and mean reciprocal rank (MRR) survive a causal index (unconditionally and on answerable queries), and does system ranking change? **RQ2:** How much of the random-vs-causal gap is answerability loss, pool-size effects, training leakage, or missing labels for earlier sibling debunks, and how does the remaining gap relate to the time lag between post and debunk, lexical overlap and narrative drift (covariates)? **RQ3:** Is the causal penalty larger for Polish than for English, and how much do earlier English fact-checks recover for Polish posts?

Hypotheses (prespecified, not formally preregistered). Most MultiClaim gold pairs are post-before-fact-check (H1); on answerable queries, time-local distractors are harder than a size-matched random pool (H2); Polish shows a larger causal penalty than English, especially for lexical BM25 (H3); fine-tuned retrievers lose more under chronological training than zero-shot ones (Shaar et al., 2022) (H4).

Whether large language model (LLM) rerankers gain on facts seen in pretraining is left to future work. Language and retrieval bias across languages is analysed by Vykopal et al. (2026) and is out of scope here; we only compare each language with itself over time. Random splits overestimate performance (Søgaard et al., 2021), which motivates the time-respecting split.

2 Significance and state of the art

We summarise fifteen prior works and, after the Phase 1 review, four more. Table 1 lists the published numbers we compare on one 0–1 scale, with the dataset and the split or evaluation con-

dition named for every value ("no split" where the source has none). No PFCR evaluation we found restricts the fact-check index *per query* to documents published before the post. A search on 11 Nov 2026 (OpenAlex, arXiv and forward citations of ten seed papers; works since 2019; queries in `research/protocols/causal-index-pfcr.yaml`) returned 1,738 records, 1,114 unique, 240 after keyword screening and 6 read at abstract level; none restricts the PFCR index per query. The closest restrict evidence to what predates the claim in verification (Lu and Setiawan, 2026) or fence citation context in case-law retrieval, whose benchmark already limits each query's candidates to earlier cases (Liu et al., 2026).

PFCR task and shared tasks. Shaar et al. (2020) define PFCR as ranking verified claims for a social-media or debate input; on Snopes tweets and PolitiFact debate sentences, BM25 plus a ranking support vector machine (rankSVM) reaches MRR 0.788 and S@10 0.915 on the Snopes test (BM25 top-50 rerank). Shaar et al. (2021) run Task 2 of CheckThat! 2021 at the Conference and Labs of the Evaluation Forum (CLEF) (English/Arabic tweets; English debates) with fixed pools and no date filter; best English tweet MAP@5 is 0.883 (Aschern) vs BM25 0.749 on the 70/15/15 tweet split. Nakov et al. (2022) rerun the lab in 2022 with a small new test set; RIET Lab reaches MAP@5 0.956 on English tweets (2022 test) with sentence-T5 plus GPT-Neo reranking. Hardalov et al. (2022) mine 333k noisy tweet–Snopes pairs from crowd replies and train sentence-BERT; on CheckThat! 2021 English test they report MAP@5 0.903 for the full system (vs Aschern 0.882 as quoted in their table; the lab overview reports 0.883).

Multilingual PFCR corpora and systems. Pikuliak et al. (2023) release MultiClaim (28k posts, 206k fact-checks, 31k pairs) and evaluate unsupervised and fine-tuned retrievers over all post–fact-check pairs with no held-out split, searching same-language fact-checks only; best monolingual S@10 is 0.83 (Generalizable T5-based dense Retriever, GTR-T5-Large, on English translations, averaged over languages with more than 100 pairs). Their post-newer vs post-older subsets score 0.8002 vs 0.8004 S@10 under the same index (no per-query date filter). Peng et al. (2025) organise SemEval-2025 Task 7 on Multi-Claim with out-of-time test posts (pool to May 2024); winner PINGAN AI reaches monolingual average S@10 0.9601 and Polish 0.9260 on the hidden test languages. Devadiga et al. (2025) (a SemEval system) translate posts, fine-tune Stella with hard negatives and LLM-rerank; monolingual test S@10 0.938 (Polish 0.886). Ramponi et al. (2025) compare multilingual vs crosslingual retrieval on a curated MultiClaim v2 subset (47 languages) with an 80/10/10 language-stratified split that separates fact-checks across folds; LLM reranking of multilingual-e5-large reaches MRR@10 0.8042 multilingual. Vykopal et al. (2025) release AMC-16K (16k human Yes/No pairs from MultiClaim; 16% relevant overall) and evaluate LLMs as relevance detectors; best Macro-F1 0.825 on the monolingual half (Mistral Large few-shot).

Closest temporal precedents for PFCR. Shaar et al. (2022) add transcript context and a debate-level chronological train/test split on PolitiFact; the prior SOTA reranker drops from MAP 0.602 (sentence-level random) to 0.429 (debate chrono) without restricting the claim pool by date.

Time-respecting IR and verification (not PFCR index filtering). Ounis et al. (2011) define the 2011 Microblog track of the Text REtrieval Conference (TREC) real-time search with a compulsory run that bans future tweets and future corpus statistics; best compulsory P@30 is 0.4551. Liška et al. (2022) evaluate question answering (QA) over a growing dated news corpus by quarter; dated dense passage retrieval (DPR) raises recall@20 on recent generated questions from 0.43 to 0.57. Glockner et al. (2022) show that a share of 0.697 of MultiFC misinformation claims have leaked fact-check evidence in search snippets (verification leakage from later fact-checks). Schlichtkrull et al. (2023) release AVeriTeC with evidence restricted to documents before the claim and temporally ordered splits; baseline verdict macro-F1 is 0.23 (gold evidence 0.49) on the development set; the test labels were not public. Mu et al. (2023) show rumour classifiers collapse under chronological splits: on Twitter 16, BERT macro-F1 falls from 0.915 (random) to 0.472 (stratified chronological).

Added after the Phase 1 review. Vykopal et al. (2026) show that dense retrievers return some

fact-checks far more often than their gold share, so a language penalty may reflect retrieval bias; we therefore report within-language changes only. Kazemi et al. (2021) frame claim matching beyond English and rerank BM25 candidates with a distilled multilingual encoder. Søgaard et al. (2021) show that random splits overestimate performance on later data, the premise of our time-respecting split. Alkhalifa et al. (2023) evaluate retrieval on dated collection snapshots, a coarser precedent for a time-respecting index.

Table metrics. Let $G(q)$ be the gold items for query q and r_q the rank of the first gold hit (undefined if none). *S@10* is the fraction of queries with a gold hit in the top 10; *MRR* is the mean of $1/r_q$ (0 if no gold ranked) (Pikuliak et al., 2023; Peng et al., 2025). *MRR@10* clips reciprocal rank at 10. *MAP@k* averages, over queries, precision at each relevant hit within the top k (plain MAP uses the full list). *P@k / R@k*: fraction of the top k that are relevant / fraction of all gold recovered in the top k. *Macro-F1*: unweighted mean of per-class F1 (classification, not ranking). *NDCG*: gain discounted by rank and normalised by the ideal ranking. *Leak rate*: share of claims whose later fact-check evidence already appears in search snippets (Glockner et al., 2022).

3 Datasets and status

Table 2 lists the source datasets and the two datasets we will create. For ranking corpora, class balance is the density of positive post–fact-check links (gold pairs per post, share of fact-checks with a linked post), not a balanced classification split. Balance is measured on the raw files (`scripts/corpus_stats.py`): gold fact-checks per post (FC/post) and the share of fact-checks with at least one linked post. Of 28,092 v1 and 89,139 v2 posts, 25,957 and 87,304 are dated and kept, the rest are dropped; 202,716 of 205,751 v1 and 430,630 of 435,252 v2 fact-checks are dated. The Polish and English rows are monolingual v2 slices (posts, fact-checks and pairs of that language); all 9,044 Polish and 166,080 of 166,430 English fact-checks are dated. The v1 files we received have no language field and an empty text column, and their post ids differ from v2, so v1 Polish and English post counts are the published ones.

4 Concept, methodology and work plan

Pipeline. Figure 1 summarises the evaluation path. Systems (committed core, four): BM25; multilingual-e5-large (Wang et al., 2024); the Polish–English retrieval encoder mmlw-retrieval-roberta-large (chosen over the Language-agnostic BERT Sentence Embedding, LaBSE, because it was trained on Polish–English pairs); bge-reranker-v2-m3 on the top 100 (Chen et al., 2024). Protocols: full-pool random; strict causal $P(q)$; size-matched random of $|P(q)|$. Compute: local open-weight models only; no commercial LLM API on post text.

Primary metric. The central claim is the Polish–English difference-in-differences of unconditional S@10 between the random and the strict causal pool (unanswerable queries count as misses). MRR and the answerable-only versions are secondary. The per-language answerable rate is the first result reported. All protocols are scored on one fixed query set per language, so the random, size-matched and causal pools differ only in the index.

First result: answerable rate. Most gold pairs are post-before-fact-check: 85% in v1, 93% in v2 and 89% of the Polish v2 pairs. Posts with an earlier gold fact-check (margin 0), as a share of posts with a gold pair: v1 15.1% and v2 7.7%; monolingual v2 Polish 152 posts (11.1%) and English 2,364 (10.5%); SemEval test Polish 33 (6.6%) and English 72 (14.5%). With margins of 1 and 7 days the Polish count falls to 14 and 9 posts and the English count to 1,058 and 567.

Proof of concept. BM25 (bm25s, original-language words) scores one fixed query set per language, the posts with a same-language gold fact-check, under the three pools (Table 3); intervals are paired bootstrap over queries (1,000 resamples), and the size-matched pool is averaged over three seeds. Unconditionally S@10 falls from 0.631 to 0.091 for Polish and from 0.553 to 0.051 for English, a Polish–English difference-in-differences of -0.037 (95% bootstrap interval $[-0.062, -0.010]$). On answerable queries the causal pool costs little: 0.829 to 0.822 for Polish (152 queries) and 0.519 to 0.492 for English (2,364), so the drop is almost all answerability; English size-matched pools are easier than causal ones (0.528 vs 0.492), in the direction of H2.

Table 1: Published numbers for the works above. All values on a 0–1 scale; each names the dataset and the split or evaluation condition of the source paper. Metrics as above.

Work	Metric	Value	Dataset / split or condition
(Shaar et al., 2020)	MRR	0.788	Snopes tweets / test; BM25 top-50 + rankSVM
(Shaar et al., 2020)	S@10	0.915	same
(Shaar et al., 2021)	MAP@5	0.883	CheckThat! 2021 English Task 2A / test of the 70-15-15 split (Aschern; 0.882 in (Hardalov et al., 2022))
(Shaar et al., 2021)	MAP@5	0.749	same / BM25 baseline
(Nakov et al., 2022)	MAP@5	0.956	CheckThat! 2022 English Task 2A / 2022 test (RIET Lab)
(Hardalov et al., 2022)	MAP@5	0.903	CheckThat! 2021 English / test; CrowdChecked-trained system
(Pikuliak et al., 2023)	S@10	0.83	MultiClaim / all pairs, no split; same-language pool; GTR-T5-L on English text
(Pikuliak et al., 2023)	S@10	0.8002 / 0.8004	MultiClaim / post-newer vs post-older, no split, same index, GTR-T5
(Peng et al., 2025)	S@10	0.9601	SemEval-2025 Task 7 / monolingual test average (PINGAN AI)
(Peng et al., 2025)	S@10	0.926	same / Polish test (PINGAN AI)
(Devadiga et al., 2025)	S@10	0.938	SemEval-2025 Task 7 / monolingual test (TIFIN)
(Devadiga et al., 2025)	S@10	0.886	same / Polish test (TIFIN)
(Ramponi et al., 2025)	MRR@10	0.804	MultiClaim v2 curated / test partition of the 80-10-10 split; LLM rerank
(Vykopal et al., 2025)	Macro-F1	0.825	AMC-16K / monolingual half, no split; Mistral Large few-shot
(Shaar et al., 2022)	MAP	0.602	PolitiFact debates / sentence-level random split
(Shaar et al., 2022)	MAP	0.429	same / debate-level chronological split
(Ounis et al., 2011)	P@30	0.455	TREC 2011 Microblog / 49 judged topics, no split; compulsory no-future run
(Liška et al., 2022)	R@20	0.57 / 0.43	StreamingQA / 2020 recent generated questions; dated vs undated DPR
(Glockner et al., 2022)	leak rate	0.697	MultiFC / all misinformation claims, no split; automatic leak detector
(Schlichtkrull et al., 2023)	macro-F1	0.23 / 0.49	AVeriTeC / development set; baseline vs gold evidence
(Mu et al., 2023)	macro-F1	0.915 / 0.472	Twitter 16 / random vs stratified chronological split, BERT
Added after the Phase 1 review			
This work	S@10	0.631 / 0.091	MultiClaim v2 Polish, 1,367 queries / random vs causal pool, BM25
This work	S@10	0.553 / 0.051	MultiClaim v2 English, 22,600 queries / random vs causal pool, BM25
(Vykopal et al., 2026)	S@10	0.829	MultiClaim test split, 1,239 pairs / all fact-checks, multilingual-e5
(Kazemi et al., 2021)	F1	0.885	claim similarity set, 2,343 pairs / 10-fold cross-validation, AdaBoost on features of a distilled multilingual encoder
(Søgaard et al., 2021)	error reduction	0.808 / 0.450	SentEval bigram shift / random split vs new Gutenberg sample, BERT
(Alkhalifa et al., 2023)	NDCG	0.3226 / 0.3359	LongEval 2023 / within-time (June 2022) vs long-term (Sept. 2022) snapshot, mean of runs

Paired permutation tests (10,000 sign flips) give $p < 0.001$ for random vs causal and for size-matched vs causal unconditionally in both languages; on answerable queries the Polish difference is not significant (a single discordant query) and the English one is ($p < 0.001$). Document-frequency statistics come from the full corpus, so the causal pool still shares future statistics. Of the 152 Polish answerable posts, 138 have their gold fact-check less than a day before the post, where time zones are undocumented, so the Polish causal result is fragile. The three runs with their tests took 5 s (Polish) and 23 min (English) on a laptop (macOS, arm64) with a peak of 1.1 GB.

Dates and ties. A fact-check or post may carry several dated instances, so we take the earliest instance, at full second precision, as its time; $P(q)$ uses the strict inequality $t_f < t_q$ (margin 0), so a fact-check enters only if it was published earlier, ties are excluded, and a same-day fact-check published earlier counts. Sensitivity runs shrink the pool by margins of 1 and 7 days ($t_f < t_q - m$) and use a conservative and a liberal bound (latest

Table 2: Source datasets. FC = fact-checks: short verified/debunked claims from fact-checking sites (the documents we retrieve).

Dataset	Size	Label definition	Class balance	Licence	How obtained	Status
MultiClaim v1	206k FC, 28k posts, 31k pairs	fact-checker gold links (ClaimReview / warnings)	1.12 FC/post; 8.2% of FC linked	restricted Zenodo record, access on request; no licence field	Zenodo 7737983	READY
MultiClaim v2	435k FC, 89k posts, 105k pairs	same + relationship types	1.18 FC/post; 17.7% linked	restricted record: researchers only, non-commercial research; EU Directive 2019/790 research exemption	Zenodo 15413169	READY
SemEval-2025 T7	train from MultiClaim; test 4.3k mono posts, 4,936 pairs (PL: 499 posts, 563 pairs)	shared-task gold for S@10	1.16 FC/post; 1.7% of the 272k test pool linked	restricted record, same terms as MultiClaim, access on request	Zenodo 14989176	READY
MultiClaimNet	85.3k FC in 30.9k clusters	same-claim cluster ids	n/a (clustering)	CC BY 4.0	Zenodo 15100352	READY
AMC-16K	16k pairs / 20 langs	human Yes/No relevance	16% relevant overall; PL mono 20.25%	CC BY-NC-SA 4.0	authors (not via the Language Data Space, LDS)	RESTRICTED; requested 11 Nov 2026, answer pending
Polish / English (v1 counts from the paper, v2 and SemEval measured)						
PL	v1 (paper): 453 posts, 6,912 FC; v2: 1,406 of 1,450 posts dated, 9,044 FC, 1,550 pairs	as above	1.13 FC/post; 13.7% linked	as above	as above	SemEval test: 499 of 500 posts dated, 8,796 FC
EN	v1 (paper): 7,307 posts, 85,814 FC; v2: 24,004 of 24,396 posts dated, 166,430 FC (166,080 dated), 28,533 pairs	as above	1.26 FC/post; 13.1% linked	as above	as above	SemEval test: 497 of 500 posts dated, 145,287 FC
Datasets we will create						
Causal split	id lists per language and protocol, kept confidential; 1,367 Polish and 22,600 English queries from v2, v1 and SemEval later	post–fact-check ids with the date rule of Sec. 4	n/a	our own; the dataset authors asked us not to publish id lists, so we release the code only	built by us, W1–W2	TO COLLECT
Failure annotations	50 queries (25 PL, 25 EN)	one code per newly failing query: answerability, lexical leakage, narrative drift, cross-lingual-only debunk, missing label, timestamp error, multimodal-only (codebook frozen before annotation; two annotators, agreement by Cohen's κ)	n/a	our own	Florek and Cwalina, by 16 Dec	TO COLLECT

vs earliest instance); we report the answerable rate and headline scores under both, and the share of queries whose pool membership flips. 100 dates per language are checked against live URLs; un-

Table 3: BM25 S@10 on monolingual MultiClaim v2 slices under three pools; brackets give 95% bootstrap intervals; size-matched is the mean of three seeds (range below 0.01). Answerable: a gold fact-check predates the post.

Language (queries; answerable)	random pool	size-matched	causal pool	causal, answerable only
Polish (1,367; 152)	0.631 [0.606, 0.656]	0.484	0.091 [0.076, 0.108]	0.822 [0.766, 0.881]
English (22,600; 2,364)	0.553 [0.547, 0.560]	0.398	0.051 [0.049, 0.054]	0.492 [0.470, 0.512]

dated posts are dropped and counted. A post whose only earlier debunk is missing from the corpus counts as unanswerable; we count such cases and report them as a source of false failures.

Metrics. Chosen before results (MultiClaim / SemEval-2025 convention (Pikuliak et al., 2023; Peng et al., 2025)). Let $G(q)$ be the gold fact-checks for query q and r_q the rank of the first gold hit ($1/r_q = 0$ if none). *S@10* is the fraction of queries with $r_q \leq 10$; *MRR* is the mean of $1/r_q$. *Answerable rate* is the share of queries with $G(q) \cap P(q) \neq \emptyset$ under the causal pool. We report both unconditionally (unanswerable queries count as misses) and on answerable queries only; intervals by paired bootstrap over queries and p-values by a paired permutation test (Urbano et al., 2019); Polish–English contrast via difference-in-differences.

Concrete success deliverables. (1) Code that rebuilds the causal split from the restricted data; id lists stay confidential at the dataset authors' request, post text is never released. (2) Shared results table: $\geq$4 systems $\times$ random vs causal $\times$ Polish and English, with bootstrap intervals; central number = extra Polish temporal cost. (3) Gap decomposition plus coded analysis of 50 newly failing queries.

Week-by-week plan. Internal work finishes before Christmas; course checkpoints: mid-term video 18 Nov, Phase 2 2 Dec, formal final 13 Jan 2027. One owner per task.

- **W1 (to 18 Nov):** date audit of 100 URLs per language (Florek).
- **W2 (to 25 Nov):** BM25 and multilingual-e5 on all three pools, PL and EN (Jarosz); mmlw and reranker, graphics-processing-unit (GPU) runs (Florek).
- **W3 (to 2 Dec):** causal and size-matched runs for all four systems (Jarosz); Phase 2 report and proof-of-concept (PoC) table (Cwalina).
- **W4 (to 9 Dec):** bootstrap intervals and difference-in-differences (Florek); gap decomposition with covariates (Pozorski).
- **W5 (to 16 Dec):** 50 failure annotations and codebook (Cwalina; Florek second annotator); judge validation (Jarosz).
- **W6 (to 23 Dec):** release check of the code (Pozorski); write-up (Cwalina).

Main risks: the Polish answerable count is 152 posts at margin 0, barely above 150, and 14 and 9 at margins of 1 and 7 days, so Polish margin sensitivity is descriptive; timestamp semantics are undocumented (URL date audit); AMC-16K was requested on 11 Nov and is pending.

5 Data statement

Primary data: MultiClaim v1/v2 and SemEval-2025 Task 7 from restricted Zenodo records (research use only: the v2 and SemEval records limit access to researchers for non-commercial research under the EU Directive 2019/790 exemption, v1 states no further terms; authors may ask for removal of their posts; no re-identification, no sharing beyond named requesters) (Moro et al., 2025; Pikuliak et al., 2023; Peng et al., 2025); MultiClaimNet (Zenodo; CC BY 4.0) (Panchendrarajan et al., 2025). Paweł Florek requested access to the Zenodo records of MultiClaim v1 and v2, SemEval-2025 Task 7 and MultiClaimNet on 3 Nov 2026 and it was approved on 4 Nov 2026. Posts are real social-media text, released in anonymised form by the dataset authors; we treat them as personal data. We store files only in a directory readable by named team members, never send post text to an external LLM API, never put raw text in git or the paper (examples paraphrased; errors aggregated), and delete the data at project end. No per-outlet or per-author error ranking will be published. AMC-16K is secondary: it validates the LLM relevance judges that pre-sort sibling fact-check candidates; failure codes are assigned

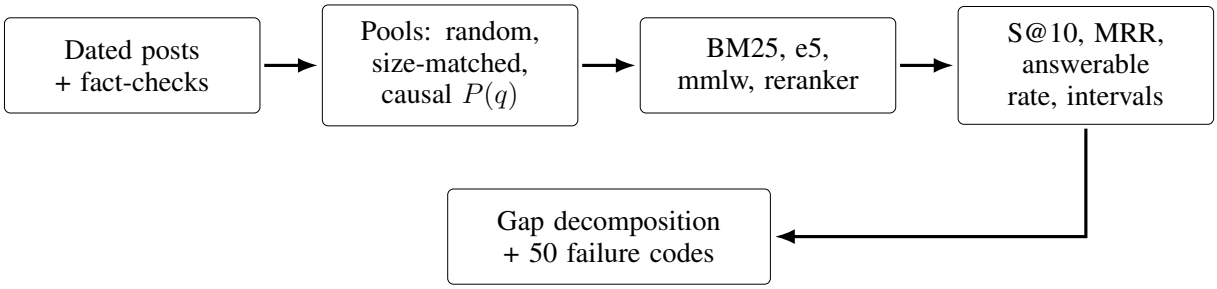


Figure 1: Evaluation pipeline.

Table 4: Contributions. Hours are approximate and cover the Phase 1 report and this revision.

Member	Named tasks	Hours
Paweł Pozorski	research plan and topic notes, paper notes in `research/`, EDEN compute notes, README, template updates; data, protocol and SOTA sections; Phase 2 revision: feedback note, response plan, rebuttal, corpus statistics, novelty search, BM25 proof of concept, permutation test	15
Mateusz Jarosz	adopted the ACL report template and the grading handbook note; data layer (MultiClaim v1/v2 and SemEval loaders, strict-causal and size-matched pools, answerable rate, exploratory data analysis (EDA) notebooks, container recipes); systems and SOTA sections	10
Paweł Florek	Phase 1 report update and timeline; evaluation section, pipeline figure and LaTeX; LLM-use and contribution sections	5
Jan Cwalina	drafting and polish of the Phase 1 report text	5

by two human annotators. Each of three open-weight judges, disjoint from the evaluated systems (gpt-oss-120b, Command A, Mistral Large 2411), gets an English prompt (post, fact-check, "Does the fact-check address the claim made in the post? Answer Yes or No.") and answers Yes or No; the label is the majority of the three. Per judge and for the majority we compute Cohen's κ against the human Yes/No labels of the 400 Polish AMC-16K pairs and use the judges only above $\kappa = 0.6$; if AMC-16K has not arrived by week 4, sibling labels are reported unvalidated. Its public listing is on the EU Language Data Space (LDS) catalogue, which our university cannot use; we requested a copy on 11 Nov 2026 from the authors at the Kempelen Institute of Intelligent Technologies (KInIT) under research terms.

6 Contribution

Table 4 lists each member's named tasks, taken from the repository history (branches `main` and `feat/data-foundation`) where it records them, and the approximate hours spent on the Phase 1 report and this revision.

7 LLM-use declaration

We used the following AI tools while preparing this proposal and report: Claude Code with Claude Sonnet 5.5 and Claude Opus 5.5 (Anthropic). In the first submission its role was limited to formatting (LaTeX tables, section structure, citation layout) and wording corrections; in this revision it also helped draft the rebuttal, run the literature search and write the data-statistics and BM25 code, and the authors reviewed these outputs.

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